THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VIII — MATHEMATICS · MID-TERM PRACTICE PAPER

Session 2026-27 | CBSE / RBSE Pattern

Class: 8 | Subject: Mathematics | Set 05 | Time: 2 Hours 30 Minutes | Max. Marks: 60

General Instructions:

1. This question paper contains 16 questions. All questions are compulsory.
2. Question paper is divided into FIVE sections – Section 'A', 'B', 'C', 'D' and 'E'.
3. In Section 'A' question number 1 has 12 multiple choice sub-parts carrying 1 mark each.
4. In Section 'B' question numbers 2 to 7 are very short answer type questions carrying 2 marks each.
5. In Section 'C' question numbers 8 to 10 are short answer (SA) type questions carrying 3 marks each.
6. In Section 'D' question numbers 11 to 13 are long answer (LA) type questions carrying 5 marks each.
7. In Section 'E' question numbers 14 to 16 are case-based questions whose sub-parts are of 1, 1 and 2 marks respectively.
8. There is no overall choice. However, internal choice has been provided in 1 question of Section B, 1 question of Section C, 2 questions of Section D, and in the 2-mark sub-part of every case-based question in Section E. Attempt only one of the alternatives in all such questions.
9. Draw neat figures wherever required. Use of calculators is NOT allowed. Write the serial number of the question before attempting it.

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) Which of the following is a perfect square? **1**

(a) 1200 (b) 1296 (c) 1250 (d) 1300

(ii) The cube root of −343 is: **1**

(a) 7 (b) −49 (c) −7 (d) −3.5

(iii) The units digit of 17^2 is: **1**

(a) 9 (b) 7 (c) 1 (d) 4

(iv) The value of $(-2)^{-3}$ is: **1**

(a) −1/8 (b) 1/8 (c) −8 (d) 8

(v) 349 in Roman numerals is: **1**

(a) CCCIL (b) CCCXLIX (c) CCXXXXIX (d) CCCXXXXIX

(vi) $(1010)_2$ in decimal is: **1**

(a) 10 (b) 8 (c) 12 (d) 5

(vii) A quadrilateral whose diagonals are equal and bisect each other at right angles is a: **1**

(a) rectangle (b) rhombus (c) kite (d) square

(viii) The diagonals of a kite meet at: **1**

(a) 60° (b) 45° (c) any angle (d) right angles

(ix) $(x - 5)(x + 5)$ equals: **1**

(a) $x^2 + 25$ (b) $x^2 - 10x + 25$ (c) $x^2 - 25$ (d) $x^2 - 10$

(x) If $x + 1/x = 3$, then $x^2 + 1/x^2$ equals: **1**

(a) 9 (b) 11 (c) 7 (d) 5

(xi) If 4 : 7 = 20 : x, then x is: **1**

(a) 28 (b) 32 (c) 40 (d) 35

(xii) 6 men build a wall in 10 days. 15 men will build it in: **1**

(a) 4 days (b) 25 days (c) 6 days (d) 9 days

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the square root of 1521 by prime factorisation. **2**

3. Find the smallest number by which 500 must be multiplied to get a perfect cube. **2**

4. Write 0.0000081 in standard form and 7.5×10^4 in usual form. **2**

5. Write the Roman numeral MCDXLVII in the Hindu-Arabic system. **2**

OR

Convert $(321)_4$ into a decimal number.

6. Two adjacent angles of a parallelogram are in the ratio 1 : 4. Find them. **2**

7. Expand $(2a - 3)^2$ and $(x + 7)(x + 3)$. **2**

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Find the cube root of −1.331 and the square root of 0.0625. **3**

9. The angles of a quadrilateral are x°, (x + 10)°, (x + 20)° and (x + 30)°. Find each angle. **3**

OR

The diagonals of rectangle ABCD meet at O. If OA = 5 cm and ∠AOB = 60°, find BD and AB.

10. Using identities, find: (i) 103^2 (ii) 59×61 (iii) $(2.1)^2$ **3**

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 2700 must be divided so that the quotient is a perfect square. Also find the square root of the quotient. **5**
(ii) Find the smallest number to be added to 1000 to make it a perfect square.

OR

(i) Find the cube root of 17576 by prime factorisation.
(ii) A cubical box has volume 5.832 m^3. Find the length of its edge.

12. Simplify: (i) $3^{-1} + 5^{-1}$ (ii) $(2^5 \times 2^{-2}) \div 2^{-1}$ (iii) $(-5)^0 \times 5^{-2}$ (iv) Write 0.00000052 in standard form (v) Find x if $2^x \div 2^{-3} = 2^8$ **5**

OR

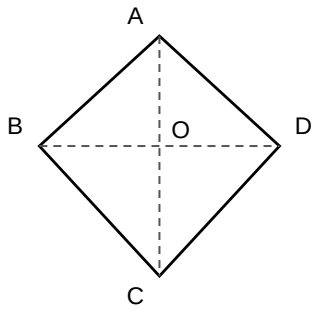
(i) Write 1994 in Roman numerals. (ii) Convert 77 to base 2. (iii) Add $(101)_2$ and $(11)_2$. (iv) Convert $(143)_5$ to decimal. (v) Convert 100 to base 3.

13. (i) Divide ₹ 5,400 among three persons in the ratio 2 : 3 : 4. **5**
(ii) If 12 notebooks cost ₹ 180, find the cost of 20 notebooks.
(iii) 3 pumps empty a tank in 8 hours. How long will 4 pumps take?
(iv) Find x if 5 : 8 :: x : 56.
(v) The ages of two brothers are in the ratio 4 : 5 and their sum is 45 years. Find their ages.

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. A kite-shaped decoration ABCD is cut from coloured paper. AB = AD = 5 cm, CB = CD = 9 cm, ∠BAD = 70° and ∠ABC = ∠ADC = 100°. The diagonals AC and BD meet at O.



(i) Which two angles of the kite are equal? 1

(ii) Find the perimeter of the decoration. 1

(iii) Find ∠BCD. 2

OR

(iii) Find ∠AOB and ∠ABO.

15. A square garden has an area of 2025 m². The owner wants to fence it on all four sides.

(i) Find the side of the garden. 1

(ii) Find the length of fencing required. 1

(iii) Find the cost of fencing at ₹ 120 per metre. 2

OR

(iii) The side is increased by 4 m. Find the area of the new square garden.

16. A rectangular hall has length (x + 7) m and width (x + 3) m, where x is a number of metres chosen by the architect. The floor is to be carpeted.

(i) Write the area of the hall as a polynomial in x. 1

(ii) Find the area when x = 5. 1

(iii) Find the cost of carpeting at ₹ 150 per m² when x = 5. 2

OR

(iii) Show that $(x + 7)(x + 3) = (x + 5)^2 - 4$.